

Separated by flooding

Earth has experienced many ice ages, when large parts of the globe were covered with ice. When temperatures increased, melting ice caused sea levels to rise. Lots of islands around the world, like Sumatra and Borneo in Indonesia, formed in this way. In the process, animals there separated and evolved into different species, including the graceful pitta in Sumatra and blue-banded pitta in Borneo.



The Sunda Shelf in Southeast Asia about 20,000 years ago

Emerging islands

When sea levels were lower, Sumatra, Borneo, and neighboring islands were part of one land mass called the Sunda Shelf. With rising sea levels, much of the region became submerged (seen in a lighter color), leaving behind islands.

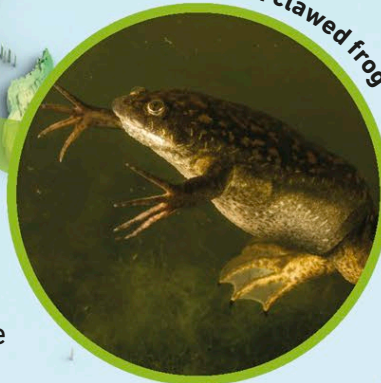
Graceful pitta



Blue-banded pitta

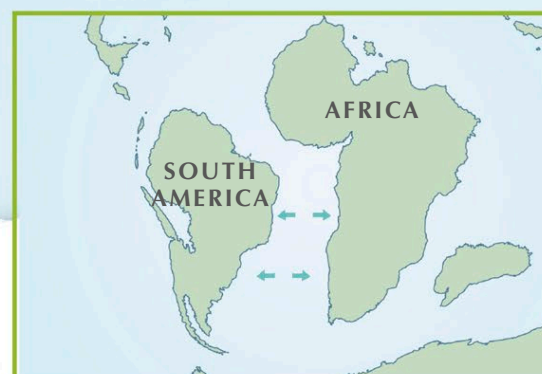


African clawed frog



Splitting up

South America and Africa were once joined, but started to drift apart 100 million years ago, separating animals. This explains why some animals are related, even with the vast Atlantic between them. One example is the Pipidae amphibians, which include South American Suriname toads and African clawed frogs.



New ocean

The matching shapes of South America and Africa is a clue that they were once joined. Volcanic activity in this area caused tectonic plates to move, pushing apart these land masses.

South America and Africa about 95 million years ago